Options + Fixed Income — Whiteboard Card

The must-memorise formulas: BS-Merton · Greeks · Black-76 · put-call parity · duration/convexity · DV01 · curve · forwards · swaps · LMM. Source: `quant_finance/CORE_FORMULAS.md` · **drill: cover the formula, write from memory, check**

◆ Universal identity (every derivative price)

$$V_0 = e^{-rT} \cdot E^Q[\ \Phi(S_T)\ ]$$

Price of any European-style derivative = discounted expectation of payoff under risk-neutral measure Q. BS is one solution (closed-form for GBM + European payoff); binomial trees, MC, PDE are others.

> **Golden rules:** pricing lives in the RISK-NEUTRAL measure (drift = r, not μ). Hedging lives in the PHYSICAL measure. Girsanov + $\Delta\theta = (\mu-r)/\sigma$ bridges them. Everything else (Heston char fn, LMM drift, LSMC bases) is reach-for.

● Black-Scholes-Merton (spot form)

d_1 / d_2

$$d_1 = [\ln(S/K) + (r - q + \tfrac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

Call / Put

$$C = S\cdot e^{-qT}\cdot N(d_1) - K\cdot e^{-rT}\cdot N(d_2)$$
$$P = K\cdot e^{-rT}\cdot N(-d_2) - S\cdot e^{-qT}\cdot N(-d_1)$$

***Inputs**: S=spot, K=strike, T=maturity (yrs), r=risk-free, q=div yield (continuous; drop for non-div stock), σ=vol. **N**=std-normal CDF. **Output**: option value at t=0 (same units as S).*

Toy example (ATM call, no divs)

$$S=100,\ K=100,\ r=0.05,\ q=0,\ \sigma=0.20,\ T=1$$
$$d_1 = [\ln(1) + (0.05 + 0.02)\cdot 1] / 0.20 = 0.35$$
$$d_2 = 0.35 - 0.20 = 0.15$$
$$N(0.35) = 0.6368,\quad N(0.15) = 0.5596$$
$$C = 100\cdot 0.6368 - 100\cdot e^{-0.05}\cdot 0.5596 = 63.68 - 53.23 = 10.45$$

Python (canonical)

```python
from math import log, sqrt, exp
from scipy.stats import norm

def bs_call(S, K, T, r, sigma, q=0.0):
    d1 = (log(S/K) + (r - q + 0.5*sigma**2)*T) / (sigma*sqrt(T))
    d2 = d1 - sigma*sqrt(T)
    return S*exp(-q*T)*norm.cdf(d1) - K*exp(-r*T)*norm.cdf(d2)

def bs_put(S, K, T, r, sigma, q=0.0):
    d1 = (log(S/K) + (r - q + 0.5*sigma**2)*T) / (sigma*sqrt(T))
    d2 = d1 - sigma*sqrt(T)
    return K*exp(-r*T)*norm.cdf(-d2) - S*exp(-q*T)*norm.cdf(-d1)
```

● Put-call parity (model-free)

$$C - P = S\cdot e^{-qT} - K\cdot e^{-rT}$$

Holds independent of any model — pure static replication (buy call + sell put + short forward = zero). Failure = arbitrage OR stale quote OR missed dividend / hard-to-borrow.

> Use to (a) derive one side from the other, (b) sanity-check pricer outputs, (c) spot data errors on live quote screens.

● Black-76 (forward form)

$$d_1 = [\ln(F/K) + \tfrac{1}{2}\sigma^2 T] / (\sigma\sqrt{T}),\quad d_2 = d_1 - \sigma\sqrt{T}$$
$$C = e^{-rT}\cdot[F\cdot N(d_1) - K\cdot N(d_2)]$$
$$P = e^{-rT}\cdot[K\cdot N(-d_2) - F\cdot N(-d_1)]$$

Use for anything quoted on a FORWARD: swaptions (F = fwd swap rate), caplets (F = fwd LIBOR), FX (F = fwd FX), commodity futures. No q needed — the forward already prices carry.

Bridge to BS-Merton: substitute $F = S\cdot e^{(r-q)T}$ into Black-76 → recover BS-Merton for spot options.

▲ Greeks (BS-Merton call)

Greek	Call formula	Meaning
Δ delta	$e^{-qT}\cdot N(d_1)$	∂C/∂S · hedge ratio
Γ gamma	$e^{-qT}\cdot\varphi(d_1) / (S\cdot\sigma\cdot\sqrt{T})$	∂²C/∂S² · convexity (same call/put)
ν vega	$S\cdot e^{-qT}\cdot\varphi(d_1)\cdot\sqrt{T}$	∂C/∂σ · vol sensitivity (same call/put)
Θ theta	$-[S\cdot e^{-qT}\cdot\varphi(d_1)\cdot\sigma]/(2\sqrt{T}) - r\cdot K\cdot e^{-rT}\cdot N(d_2) + q\cdot S\cdot e^{-qT}\cdot N(d_1)$	∂C/∂t · time decay (usually neg for long option)
ρ rho	$K\cdot T\cdot e^{-rT}\cdot N(d_2)$	∂C/∂r · rate sensitivity

φ = std-normal PDF. **Put signs**: $\Delta_{put} = \Delta_{call} - e^{-qT}$; $\rho_{put} = -K\cdot T\cdot e^{-rT}\cdot N(-d_2)$. Γ, ν identical for call/put.

Toy Greeks (ATM call, S=K=100, r=0.05, σ=0.20, T=1)

$$\Delta \approx N(0.35) = 0.637$$
$$\Gamma \approx \varphi(0.35)/(100\cdot 0.20\cdot 1) = 0.0188$$
$$\nu \approx 100\cdot\varphi(0.35)\cdot 1 = 37.52\quad\text{(per unit σ; per 1 vol pt ≈ 0.375)}$$
$$\theta \approx -6.4/\text{year}\quad\text{(calendar; −6.4/252 ≈ −0.025/trading day)}$$
$$\rho \approx 100\cdot 1\cdot e^{-0.05}\cdot N(0.15) = 53.2\quad\text{(per 100\% rate; per 1bp ≈ 0.0053)}$$

◆ Risk-neutral pricing (why BS works)

$$V_0 = e^{-rT}\cdot E^Q[\Phi(S_T)]\quad\leftarrow\text{universal identity}$$
$$\text{Under Q:}\quad dS = r\cdot S\cdot dt + \sigma\cdot S\cdot dW_Q\quad\leftarrow\text{drift = r (not μ)}$$
$$\text{Girsanov: }\theta = (\mu - r)/\sigma\quad\text{bridges } P\leftrightarrow Q$$

Under measure Q, the discounted price of any tradable asset is a martingale. BS falls out for GBM + European payoff. Binomial trees replicate this identity in discrete time; MC estimates the E^Q directly.

◆ Bond pricing

Discrete-compound price

$$P = \Sigma_t\ CF_t / (1+y)^t$$

***CF_t** = cashflow at time t (coupons + principal). **y** = YTM (single discount rate). **Output**: dirty price.*

Continuous compound

$$P = \Sigma_t\ CF_t\cdot\exp(-y\cdot t)$$

Toy example (5y 4% annual coupon, y=5%)

$$\text{CFs: } 4,\ 4,\ 4,\ 4,\ 104$$
$$P = 4/1.05 + 4/1.05^2 + 4/1.05^3 + 4/1.05^4 + 104/1.05^5$$
$$= 3.810 + 3.628 + 3.455 + 3.291 + 81.487$$
$$= 95.67\quad\text{(trades at discount — y > coupon)}$$

◆ Duration + convexity + DV01

Macaulay + Modified duration

$$D_{Mac} = (1/P)\cdot\Sigma_t\ t\cdot CF_t/(1+y)^t$$
$$D_{Mod} = D_{Mac} / (1+y)$$

D_Mac = weighted average time to CF (years). D_Mod = price elasticity to yield. Interpretation: $\partial P/\partial y \approx -D_{Mod}\cdot P$.

Convexity

$$Conv = (1/P)\cdot\Sigma_t\ t(t+1)\cdot CF_t / (1+y)^{t+2}$$

Price change (2nd-order Taylor)

$$\Delta P/P\ \approx\ -D_{Mod}\cdot\Delta y\ +\ \tfrac{1}{2}\cdot Conv\cdot(\Delta y)^2$$

First term is duration hedge; second is the convexity "kick" that saves you when yields move a lot (both directions).

DV01 (dollar value of 1bp)

$$DV01 = D_{Mod}\cdot P\cdot 0.0001$$

\$ P&L from a +1bp parallel curve move. For the toy 5y bond above at y=5%: D_Mac = 4.62y, D_Mod = D_Mac/1.05 = 4.40y, DV01 = 4.40·95.67·0.0001 ≈ \$0.042 per \$100 face.

● Forward rate + LIBOR product

Forward rate (continuous)

$$f(t_1, t_2) = (1/(t_2-t_1))\cdot\ln[\ P(0,t_1) / P(0,t_2)\ ]$$

***P(0,T)** = today's price of a zero-coupon bond paying \$1 at T. **$f(t_1,t_2)$** = the fair rate agreed today for a loan running from t_1 to t_2. Directly bootstrappable from a discount curve.*

Bond from forward LIBORs (telescoping product)

$$P(t, T_n) = P(t, T_0)\cdot\Pi_{i=0..n-1}\ 1 / (1 + \delta_i\cdot L_i(t))$$
$$L_i(t) = (1/\delta_i)\cdot[\ P(t,T_i)/P(t,T_{i+1}) - 1\ ]$$

*Discrete analogue of $P(t,T) = \exp(-\int r\ du)$. **$\delta_i$** = accrual (yr fraction). Used everywhere: swap pricing, caplet payoffs, LMM simulator, curve construction.*

Toy: 6M forward from 6M+12M ZCB

$$P(0,\ 0.5) = 0.9754\quad\text{(6M ZCB)}$$
$$P(0,\ 1.0) = 0.9512\quad\text{(12M ZCB)}$$
$$f(0.5,\ 1.0) = (1/0.5)\cdot\ln(0.9754/0.9512) = 5.02\%$$
$$\text{(annual simple } L = (P(0.5)/P(1.0) - 1)/0.5 = 5.09\%)$$

● Par swap rate

$$S = [\ P(0,T_0) - P(0,T_n)\ ] / [\ \Sigma_{i=1..n}\ \tau_i\cdot P(0,T_i)\ ]$$

Numerator = PV of floating leg (telescopes to first ZCB − last ZCB via LIBOR product identity). **Denominator** = fixed leg PV01 (annuity). **S** is the fixed rate that sets PV(swap) = 0 at inception.

Toy: 2y annual swap

$$P(0,1) = 0.9524,\quad P(0,2) = 0.9070$$
$$S = (1 - 0.9070) / (1\cdot 0.9524 + 1\cdot 0.9070)$$
$$= 0.0930 / 1.8594 = 5.00\%$$

▲ LIBOR Market Model (LMM/BGM)

$$dL_i(t) = \sigma_i(t)\cdot L_i(t)\cdot dW_i^{T_{i+1}}(t),\quad i = 0..N-1$$

Each forward LIBOR L_i is **lognormal under its own T_{i+1}-forward measure**. So caplets price by Black-76 with vol σ_i for free — the model is calibrated to the cap-vol surface by construction.

> **Drift correction** when switching to a common measure (terminal Q^{T_N} or spot Q^B) is derived on the whiteboard from Girsanov + numeraire change. Don't memorise index ranges + signs — derive them.

Calibration workflow: (1) bootstrap discount curve from LIBOR + swap quotes. (2) strip caplet vols from cap prices via Black-76. (3) fit $\sigma_i(t)$ to caplet vols. (4) simulate under chosen measure with drift correction. (5) reprice + verify.

▲ Implied vol — inversion + ATM approx

$$\text{Solve } \sigma \text{ s.t. } BS(\sigma) = \text{market\_price}$$
$$\text{Numerical: Brent's method on } f(\sigma)=BS(\sigma)-mkt,\ \sigma\in[1e{-}6,\ 5]$$
$$\text{Brenner-Subrahmanyam ATM approx:}$$
$$\sigma_{ATM}\approx (C\cdot\sqrt{2\pi/T}) / S\quad\text{(call, r=q=0)}$$
$$\text{(accurate to \textasciitilde1\% for near-ATM; use as starting guess)}$$

Inputs: market call price C, spot S, maturity T. **Output**: implied vol. Use for a warm-start on Newton/Brent, or a quick sanity check ("does this quoted price imply a sensible vol?").

◆ Rates-desk canonical workflow

1. **Discount curve** — bootstrap from deposits (short) + FRA (middle) + IRS (long) → $P(0, T_i)$ at each pillar
2. **Forwards** — $L_i = (P(0,T_i)/P(0,T_{i+1}) - 1) / \delta_i$ (simple) or continuous compound
3. **Par swap** — $S = (P_0 - P_n) / \Sigma\ \tau_i\cdot P(0,T_i)$
4. **Swap PV** — $PV = N\cdot[c\cdot A - (1 - P_n)]$ where c = fixed rate, A = annuity factor
5. **DV01** — bump curve +1bp, reprice, ΔPV
6. **Vega/Caplet** — Black-76 on each fwd L_i with caplet vol → sum for cap

◆ Reach-for (know NAME, look up formula)

Topic	When it matters
SVI parameterisation (Gatheral 2004)	fit vol surface for exotic pricing
Heston characteristic function (Lewis 2001)	stoch-vol pricing via FFT
Carr-Madan FFT	fast options pricing when char fn known
SABR Hagan approx	swaption/cap smile fit
Bachelier (normal model)	near-zero / negative forwards (rates '20-22, oil crash)
Brenner-Subrahmanyam IV approx	near-ATM IV closed form
Longstaff-Schwartz (LSMC)	American options via MC + regression
Black-Litterman	portfolio views + prior blend
Nelson-Siegel curve fit	smooth yield curve param
LMM drift under Q^B / Q^{T_N}	path simulation — derive from Girsanov
Rebonato correlation param	LMM inter-forward corr

▲ Top traps (6-month memory)

- **Drift = r, not μ** in the pricing formulas — you're always under Q. μ is only for hedging / simulation P&L.
- **q = 0 for non-div stocks** — but not zero for FX (q = foreign rate), futures (q = r), or continuous-div ETFs.
- **Black-76 vs BS-Merton** — use B76 when you have a FORWARD (swaption, caplet, futures, FX). Use BS-M when you have SPOT + div-yield.
- **d_1 vs d_2 mix-up** — call price uses $N(d_1)$ for the S term and $N(d_2)$ for the K term. Remember: $d_2 < d_1$; both flip sign for put.

- **YTM ≠ real interest rate** — YTM is the single rate that prices the given CF stream. Different bonds with same maturity CAN have different YTMs.
- **Dirty vs clean price** — clean = dirty − accrued; quoted is clean, settled is dirty.
- **Duration signs** — D_Mod is positive; the effect on P is $-D_{Mod}\cdot\Delta y$ (rate up → price down).
- **Convexity gift** — always positive for vanilla bonds; it means D_Mod overstates loss when yields RISE and understates gain when yields FALL.
- **DV01 units** — dollars per +1bp move. "PV01" is the same thing for a swap notionalised at $1.
- **Forward rate is FORWARD-LOOKING** — $f(t_1,t_2)$ is agreed today for a loan starting at t_1. Not the actual rate that will be observed.
- **LMM index-range signs** — never memorise; derive from Girsanov every time. Wrong drift = biased simulation.
- **Cap-vol vs swaption-vol** — different products, different smiles. Same underlying rates but different measures.
- **Greeks blow up near expiry** — $\Gamma \to \infty$ ATM as $T \to 0$; $\Theta \to -\infty$. Rebalance hedge more frequently as T shrinks.
- **Vega convention** — usually per 1 vol pt (0.01 abs) but sometimes per 1% (0.01 relative). Check your pricer's convention.
- **Put-call parity broken** = missed div OR hard-to-borrow OR stale quote OR discrete divs vs continuous q assumption.

◆ Quick-reference (recall in 5 seconds)

What	Formula
BS d_1	$[\ln(S/K) + (r-q+\tfrac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$
BS d_2	$d_1 - \sigma\sqrt{T}$
BS Call	$S\cdot e^{-qT}\cdot N(d_1) - K\cdot e^{-rT}\cdot N(d_2)$
BS Put	$K\cdot e^{-rT}\cdot N(-d_2) - S\cdot e^{-qT}\cdot N(-d_1)$
Put-call parity	$C - P = S\cdot e^{-qT} - K\cdot e^{-rT}$
Black-76 call	$e^{-rT}\cdot[F\cdot N(d_1) - K\cdot N(d_2)]$, $d_1 = [\ln(F/K)+\tfrac{1}{2}\sigma^2 T]/(\sigma\sqrt{T})$
Delta (call)	$e^{-qT}\cdot N(d_1)$
Gamma	$e^{-qT}\cdot\varphi(d_1) / (S\cdot\sigma\cdot\sqrt{T})$
Vega	$S\cdot e^{-qT}\cdot\varphi(d_1)\cdot\sqrt{T}$
Risk-neutral pricing	$V_0 = e^{-rT}\cdot E^Q[\text{payoff}]$
Bond price	$P = \Sigma\, CF_t/(1+y)^t$
Macaulay duration	$D_{Mac} = (1/P)\cdot\Sigma\, t\cdot CF_t/(1+y)^t$
Modified duration	$D_{Mod} = D_{Mac} / (1+y)$
Convexity	$Conv = (1/P)\cdot\Sigma\, t(t+1)\cdot CF_t/(1+y)^{t+2}$
Price change	$\Delta P/P \approx -D_{Mod}\cdot\Delta y + \tfrac{1}{2}\cdot Conv\cdot(\Delta y)^2$
DV01	$D_{Mod}\cdot P\cdot 0.0001$
Forward rate	$f(t_1,t_2) = \ln(P(t_1)/P(t_2)) / (t_2-t_1)$
LIBOR product	$P(0,T_n) = P(0,T_0)\cdot\Pi\,(1+\delta_i\cdot L_i)^{-1}$
Par swap	$S = (P_0-P_n) / \Sigma\,\tau_i\cdot P(0,T_i)$
LMM spec	$dL_i = \sigma_i\cdot L_i\cdot dW_i^{T_{i+1}}$ (each fwd lognormal in own T-fwd measure)

◆ Daily drill protocol (10-15 min)

1. Open a blank cell.
2. Pick one section (rotate: Options / Curve / Bond / Swap / LMM).
3. Write each formula from memory.
4. Check against this card.
5. For ones you missed, also implement in code (write `bs_call(S,K,T,r,σ)` from scratch — no peeking).
6. Whole card drilled in a week; whole set retained if drilled daily.

> **Retrieval is the only thing that cements; re-reading is theatre.** If you can derive it, the formula is just bookkeeping. If you can't, work through the notebook that owns it.